

Article

Generative Artificial Intelligence Amplifies the Role of Critical Thinking Skills and Reduces Reliance on Prior Knowledge While Promoting In-Depth Learning

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Abstract: This study examines how generative AI (GAI) impacts primary students' in-depth learning, focusing on critical thinking and prior knowledge. A quasi-experiment involved 163 sixth-graders divided into three groups: a control group (lecture-based instruction) and two experimental groups using GAI as a cognitive tool (materials generation) or thinking tool (critical analysis), in which 126 participants successfully completed all the tests and were included in the analysis. ANOVA revealed the thinking-tool group and cognitive-tool group both outperformed the control group in in-depth learning, which was reflected by the knowledge transfer. Hierarchical regression showed students' critical thinking skills and use of generative artificial intelligence significantly contributed to their in-depth learning, while prior knowledge did not. Further analysis found that significant interaction effects existed between the use of generative artificial intelligence and critical thinking skills, while no significant interaction was found between the use of generative artificial intelligence and students' prior knowledge. In sum, critical thinking amplified GAI's impact, while prior knowledge showed no interaction. The results suggest GAI enhances deep learning when integrated with critical thinking, reducing reliance on prior knowledge. Educators should prioritize fostering critical thinking to maximize GAI's benefits. The findings underscore the need for pedagogical designs that balance GAI's cognitive support with metacognitive skill development.

Keywords: generative artificial intelligence; in-depth learning; critical thinking skills; prior knowledge; cognitive tool; thinking tool



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1. Introduction

In-depth learning is an essential competency and lifelong learning skill that engages learners in the process through which they can synthesize, analyze, evaluate, and apply what is learned in one situation to new situations (National Research Council, 2012; Marton & Säljö, 1976). Numerous studies have explored ways of enhancing in-depth learning through technology and tools (Aderibigbe, 2021; Cai & Gu, 2019; Sugden et al., 2021). In recent years, the technological advancements exemplified by generative artificial intelligence (GAI) have sent shock waves to education, reshaped its landscape, and offered new opportunities and challenges (Baidoo-Anu & Ansah, 2023; Milano et al., 2023). GAI has triggered increased attention in education for its potential to assist instruction and learning in various ways. However, critical questions regarding integrating GAI to promote

in-depth learning have become more prevalent in education (Perera Molligoda Arachchige, 2023; Yan et al., 2024).

Information technology tools supporting learning can be classified into cognitive and thinking tools (Zhao et al., 2022). Cognitive tools focus more on extending learners' cognition and sharing lower-order cognition to free up mental resources for learners' higher-order cognitive processing through human-machine interactions, while thinking tools provide learners with thinking strategies so that they can process information more strategically and finally internalize these strategies into their mindset (Zhao et al., 2022). As a new emerging-information technology, GAI can be designed as a cognitive or thinking tool. However, research integrating GAI into learning as a cognitive or thinking tool and comparing its effects is still scarce. We still do not know if GAI can help with in-depth learning. If it does, we are unsure whether GAI has the most influence, or if other factors like prior knowledge or critical thinking skills are more important. We also do not know how GAI affects the roles of prior knowledge and critical thinking skills in in-depth learning. Therefore, this study aimed to bridge this gap by integrating GAI as a thinking and cognitive tool to facilitate students' in-depth learning.

Existing research revealed that learners' factors, such as prior knowledge, thinking skills, and age, impacted the effectiveness of facilitating learning by information technology (Bonsaksen, 2018; Wang et al., 2020). According to the meaningful learning theory, learners' prior knowledge is the most important single factor that influences their learning (Ausubel, 1963). Critical thinking is also crucial to in-depth learning effectiveness for it inevitably influences the use of technology (Gökçearslan et al., 2019). As the accuracy of the information produced by GAI remains to be enhanced (Bhattacharyya et al., 2023), learners must approach the responses of GAI with caution, utilizing their prior knowledge and critical thinking to assess the information's correctness.

Therefore, this study aimed to explore the effectiveness of integrating GAI into Information and Communication Technology (ICT) courses as a thinking and cognitive tool for in-depth learning and the influencing variables that affect the use of GAI to promote in-depth learning. Considering critical thinking skills have been reviewed to be a stable competency that could not be promoted in a short period (P. A. Facione, 1990), different from the existing research viewing critical skills as a dependent variable, this study explored the role of critical thinking skills as a predictor that might potentially influence the effect of GAI on in-depth learning. Through this examination, this study aimed to contribute to a deeper understanding of students' in-depth learning in the context of GAI integration, shedding light on the role of its usage in the K-12 context.

2. Literature Review

2.1. In-Depth Learning

In-depth learning, also often referred to as deep learning in the context of education, is a term that encompasses various concepts related to profound understanding and engagement with learning material. Rather than just memorizing facts, Marton and Säljö (1976) emphasized understanding the underlying principles and concepts and defined in-depth learning as "involving students in activities that require a high level of cognitive processing such as synthesis, analysis, and evaluation" from the depth of approaches to learning. From the perspective of 21st-century skills, in-depth learning involves students developing a complex set of skills that include critical thinking, problem-solving, communication, and collaboration, enabling them to apply their knowledge in authentic, real-world contexts (National Research Council, 2012). These definitions reflect the multifaceted nature of in-depth learning, all emphasizing cognitive engagement, understanding, knowledge application, and higher-order thinking skills development. In this study, we define in-depth

learning as “the process by which individuals become able to apply what they have learned in one context to new contexts (i.e., transfer)”.

The evaluation system for in-depth learning has continuously evolved since the development of the Approaches to Study Inventory (ASI) questionnaire by [Entwistle and Ramsden \(1983\)](#), which aimed to measure students’ learning intentions and methods. Subsequently, [Biggs and Collis \(1989\)](#) developed the SOLO model to evaluate in-depth learning by classifying learners’ cognitive complexity into prestructural, unistructural, multistructural, relational, and extended abstract structural. In addition, based on the Presage–Process–Product (3P) model, [Biggs et al. \(2001\)](#) further developed a revised two-factor version of the Study Process Questionnaire assessing deep and surface learning approaches from the individual, environment, and task dimensions. These evaluation approaches all emphasize the importance of knowledge transfer, which is a crucial indicator of in-depth learning.

2.2. GAI and Its Use in Education

Since December 2022, the rapid development of GAI, such as ChatGPT, ERNIE Bot, and Kimi, has ushered AI development into a new climax. As a conversational AI that employs natural language processing, GAI can better understand natural language and generate new content based on trained data, bringing new opportunities and challenges to the field of education ([Kasneci et al., 2023](#)).

The development of GAI has triggered attention in education for its advantages, such as self-learning, knowledge acquisition and application, personalization, and novelty ([Jo, 2024](#)). Various studies have focused on the potential of introducing GAI in education. For example, [Su et al. \(2023\)](#) indicated that ChatGPT 3.5, a well-known GAI tool, could act as an effective scaffold in argumentative writing classes by providing support in terms of outline and language and offering personalized feedback. [Adiguzel et al. \(2023\)](#) suggested that GAI would improve teaching efficiency, support personalized teaching, promote teacher professional development, and reduce the workload of teachers and educational administrators. [Essel et al. \(2022\)](#) found that students who interacted with GAI for learning performed better academically than those who interacted with teachers. GAI has also been indicated to potentially support the cultivation of creative thinking and problem-solving abilities ([Habib et al., 2024](#); [Urban et al., 2024](#)).

GAI’s role as a tool in instruction can be divided into cognitive and thinking tools ([Zhao et al., 2022](#)). [Lajoie and Derry \(1993\)](#) defined cognitive tools as mental models or computational devices that support, guide, and expand users’ cognitive processes. [Jonassen \(1995\)](#) further emphasized the application of cognitive tools in teaching, viewing them as intellectual partners for learners, helping them to expand their thinking and become knowledge constructors. They refer to the tools that enable learners to explicitly articulate their logical and structural comprehension of various concepts ([Cai & Gu, 2019](#)), subsequently involving them in processes of critical thinking, knowledge representation, and constructing meaning ([Drew, 2019](#); [Ge et al., 2019](#)).

Thinking tools are a higher-level subset of cognitive tools. From the perspectives of theoretical basis, definition, action mechanism, and essential characteristics, [Zhao et al. \(2022\)](#) compared the differences between the ways that information technology serves as cognitive and thinking tools while being utilized to enhance instruction and learning. The differences between them are listed in Table 1.

Information technology as a cognitive tool ([Collins & Knoetze, 2014](#); [Liu et al., 2009](#)) and a thinking tool ([Koo & Seo, 2012](#); [Long et al., 2020](#)) can both have a positive impact on the final learning outcomes. GAI, as a promising information technology, can also be designed as a cognitive or thinking tool. Based on the comparison of the two tools in

Table 1, we can design different roles for GAI as the two tools. As a cognitive tool, GAI can perform information retrieval, organization, integration, and content generation, helping learners focus more of their limited cognitive resources on higher-order cognitive activities. As a thinking tool, GAI can provide thinking strategies or methods for learners in specific contexts, guiding or expanding their internal meaning construction. Table 2 summarizes GAI’s roles as cognitive and thinking tools in instruction.

Table 1. Comparison of ICT as cognitive tools and thinking tools (Zhao et al., 2022).

Dimensions	ICT as Cognitive Tools	ICT as Thinking Tools
Theoretical basis	Social constructivism theory and distributed cognition theory	Cognitive constructivism theory
Definition	Mental or computational devices that support, guide, or expand the cognitive processes of users	Thinking strategies or methods used to guide the direction and focus of users’ thinking
Action mechanism	Reducing lower-level cognitive load through external tools, allowing learners to allocate more cognitive resources to higher-order cognitive activities, creating conditions for in-depth thinking by learners	Providing thinking strategies or methods to guide or expand the internal meaning construction of learners, directly enhancing the learners’ thinking abilities
Essential characteristics	Knowledge representation, interactivity, and distributed cognition	Internalizability of thinking strategies carried by tools
Examples	Interactive Simulations (PhET): allowing students to experiment with scientific concepts virtually, represent abstract knowledge in a concrete way, and make abstract concepts more tangible	Problem Decomposition Method: guiding students to break down complex problems into smaller, manageable parts. It involves defining the core issue, decomposing it logically, assigning priorities, and solving each part step-by-step before integrating the solutions.

Table 2. GAI as cognitive tools or thinking tools.

GAI as Cognitive Tools	GAI as Thinking Tools
• Language understanding: Natural language processing • Information retrieval and integration: Retrieve and integrate large amounts of information quickly • Content generation: Generate text, answers, explanations, and more • Information organization and sorting out: Organize and sort out information to form a structured knowledge system	• Question raising: Help learners raise deeper, challenging questions • Provide thinking strategies: Provide thinking strategies to solve problems • Concept extension: Guide learners to explore related concepts • Guiding thinking process: Guide students to think according to a certain thinking path

Although GAI is posited to support in-depth learning, it does not guarantee high student achievement, especially in in-depth learning (Baidoo-Anu & Ansah, 2023; Kieser et al., 2023). Its effectiveness might be influenced by how it is used and varies across subject domains (Lo, 2023). Existing research has explored the application of improved ChatGPT-based tools in education. These tools, by simulating Socratic questioning or providing instructional interactions rather than giving answers directly, have been shown to more effectively promote students’ development in areas such as knowledge construction, self-regulated learning, and higher-order thinking skills, compared to traditional ChatGPT

(Lee, 2024; Wu et al., 2024). However, empirical evidence substantiating this claim and examining the distinct roles of GAI as a cognitive or thinking tool is also limited. Thus, further empirical evidence is needed.

2.3. Variables Influencing the Effect of In-Depth Learning in the GAI Context

In-depth learning is a complex cognitive process influenced by a variety of factors, which can be summarized into three categories: (a) learner characteristics, such as the learner's age, gender, prior knowledge, self-efficacy beliefs, learning skills, and self-assessment abilities (Bonsaksen, 2018), together with the learner's reflective behavior and personal learning effort (Wang et al., 2020); (b) learning environment, including the instructional model, the availability of learning materials, the knowledge processing level, learning effectiveness assessment, and the reflective evaluation level (Li et al., 2018); and (c) peer interaction, such as peer collaboration.

In the context of GAI-enhanced learning, prior knowledge facilitates knowledge transfer by bridging existing and emerging cognitive frameworks, and critical thinking stimulates profound cognitive reflection and innovation, thereby compensating for potential superficial learning outcomes associated with GAI utilization. However, Simonsmeier et al. (2022) conducted a comprehensive meta-analysis encompassing 8776 effect sizes, which revealed significant variability in knowledge transfer efficacy. This empirical evidence challenges universal assertions such as “knowledge is power” while simultaneously refuting claims that “the impact of prior knowledge is negligible”. These findings underscore the necessity for a systematic investigation into the effects of prior knowledge on learning outcomes while using GAI to facilitate in-depth learning.

2.3.1. Prior Knowledge

Prior knowledge significantly affects learning outcomes. F. J. R. C. Dochy et al.'s (1996) research indicates that the primary source of individual differences among students is their prior knowledge. Through a review of 183 studies, F. Dochy et al. (1999) further found that 95% of the research conclusions support the positive impact of prior knowledge on student performance. Students lacking relevant prior knowledge struggle to learn new information (O'Donnell & Dansereau, 2000). In contrast, learners with a high level of prior knowledge can significantly reduce the cognitive load in the learning process (Wilson et al., 2019). In addition, prior knowledge affects learning outcomes and learners' confidence and creativity (Ineson et al., 2013).

In the context of the digital age, the information students obtain is often fragmented, increasing the difficulty of their understanding and mastery of information. Research indicates that the effectiveness of interactive computer simulations is influenced by the user's level of prior knowledge (Park et al., 2009). Students with higher levels of prior knowledge achieve better learning outcomes when using interactive technology. In contrast, students with less prior knowledge may benefit from less interactive technology because it reduces cognitive load. Shoufan (2023) found that students without prior knowledge who used ChatGPT to answer test questions generally produced answers of slightly lower quality compared to students with prior knowledge. This indicates that ChatGPT cannot fully replace students' learning and understanding of specific topics. Students with prior knowledge can better comprehend the answers provided by ChatGPT, assess their accuracy, and make corrections when necessary. In contrast, students without prior knowledge may struggle to evaluate the reliability of ChatGPT's answers, potentially accepting inaccurate or irrelevant information. Based on this, the study aimed to explore whether the influence of prior knowledge on students' in-depth learning performance changed after the introduction of GAI.

2.3.2. Critical Thinking

Critical thinking is one of the key competencies in the 21st-century skills framework and is essential for forming and developing individuals (Dwyer et al., 2014). Critical thinking refers to the ability of learners to independently think about and evaluate information and viewpoints during the learning process, including cognitive processes such as questioning, analyzing, reasoning, and judging (Dwyer et al., 2014). It can promote learners' autonomous acquisition of knowledge, enhance their ability to analyze problems and make decisions, and improve their information processing and innovative thinking skills (Savchenko et al., 2020), crucial for in-depth learning.

Critical thinking encompasses critical thinking skills and the disposition toward critical thinking (Ennis, 1985; Halpern, 1998). Critical thinking skills are higher-order cognitive skills that require judgment, analysis, and synthesis and cannot be memorized or mechanically applied. The learning and application of critical thinking skills require specific domain knowledge (Willingham, 2008). Critical thinking disposition is an individual's intrinsic motivation to complete complex tasks, which does not need to be combined with specific domain knowledge in general contexts (Lai et al., 2011).

Existing studies identify critical thinking as the result of in-depth learning after technological intervention (Bu et al., 2023), even though some researchers argued that using technologies, such as GAI tools, might weaken critical thinking and problem-solving abilities (Dergaa et al., 2023; Johnson, 2023). However, as an essential component of information literacy, critical thinking inevitably has a certain degree of influence on the use of technology (Gökçearslan et al., 2019). Individuals with higher critical thinking skills are more equipped with cognitive skills such as analysis, evaluation, and reasoning and tend to cognitive processing of external information (N. C. Facione et al., 1994). Learners with a higher critical thinking disposition also generally have a higher level of curiosity, making them more likely to seek the truth by obtaining information from multiple sources and providing feedback (Fisher, 2001; Noone & Seery, 2018).

Considering that the accuracy of the information provided by GAI still needs improvement (Bhattacharyya et al., 2023), learners must approach the responses provided by GAI with caution, which requires applying critical thinking to discern the information. In this sense, the integration of GAI might put forward a higher requirement of learners' critical thinking skills. Therefore, this study considers critical thinking skills as an influencing factor for learners' application of GAI-integrated learning.

2.4. GAI for In-Depth Learning and Related Skills

The recent literature highlights the transformative yet complex role of GAI in fostering in-depth learning and critical thinking skills. Sardi et al. (2025) conducted a systematic review indicating that 71.4% of studies identified AI's positive impact on self-regulated learning (SRL), primarily via personalized learning, metacognitive support, and adaptive feedback. Additionally, 62.5% of studies highlighted AI's significant contribution to critical thinking, facilitating analysis, evaluation, and reflection. The researchers also warned against excessive reliance on technology, which could impair students' independent thinking (Darvishi et al., 2024; Krause et al., 2025).

2.5. The Present Study

The existing literature tells us that whether GAI can facilitate in-depth learning and critical thinking skills largely depends on how it is used. Therefore, this study proposes the following research questions:

RQ1: Did GAI have a significant effect on students' in-depth learning performance with the role of a cognitive tool and thinking tool?

RQ2: Among the factors that potentially influence in-depth learning (use of GAI, prior knowledge, critical thinking skills), which factors would have a significant impact on students' in-depth learning performance?

RQ3: Did interactive effects exist among the factors that would significantly influence students' in-depth learning performance?

3. Methodology

3.1. Context and Participants

An Information and Communication Technology (ICT) course for three sixth-grade classes in a primary school in Zhuhai, a city in the Guangdong–Hong Kong–Macao Greater Bay Area, was selected as the research context. As a top school in the city, this school is equipped with advanced information technology infrastructure, providing excellent conditions for applying GAI in learning. The teachers are also highly motivated to engage in AI-integrated teaching and learning initiatives. The ICT course refers specifically to the Information and Communication Technology (ICT) course in K-12 schools, which is primarily skill-based. In this study, the ICT course was designed to equip primary school students with fundamental digital literacy skills, such as basic computer operations, Internet usage, and simple coding concepts. Unlike content-based courses that focus on imparting specific subject knowledge or subject-specific courses that target particular areas such as Chinese literature reading and second language acquisition, the ICT course emphasizes the development of digital competencies that are essential for students' future learning and careers. It is structured to foster students' capability to effectively use digital tools for learning and communication. This course also emphasizes cultivating students' computational thinking, requiring students to think about problems systematically and logically, when participating in student-centered active learning activities. The Information Encoding unit, including the basic knowledge of data encoding, was chosen as the teaching content. This unit is proposed to enable students to represent information with numbers, letters, or text and understand that data encoding is the scientific foundation for maintaining social organization and information order. The students learned this unit via problem-based learning, in which they solved an authentic problem on school uniform encoding for lost and found.

A total of 163 students from three sixth-grade classes taught by the same teacher were recruited as the participants. The teacher had received training from the researchers on how to use GAI tools in teaching prior to the implementation of the interventions of this study. The sixth-grade students had already been exposed to digital encoding in a third-grade mathematics course and had a particular understanding of standard encoding methods such as identity card numbers. According to Piaget's theory of cognitive development (Piaget, 1971), sixth-grade students with an average age of 11, ranging from 11 to 12, are in a critical period of transition from the concrete operational stage to the formal operational stage, and they possess preliminary abstract thinking abilities. As students from this top school in the city, they had rich experience in learning via digital tools and engaging in student-centered learning activities, such as discussion, collaboration, and problem-solving. They had the capability to discover and solve problems and form concepts. However, sixth graders who employ GAI without any knowledge or critical assessment remain vulnerable.

The three classes were randomly assigned to be the Control Group (N = 54) and two experimental groups, labeled Experimental Group 1 (N = 54) and Experimental Group 2 (N = 55). The Control Group received non-GAI-involved instruction. The same instructional input was provided to the three groups, and the only variation was the usage of the GAI tool. Experimental Group 1 utilized GAI as a cognitive tool for instructional intervention, and Experimental Group 2 employed GAI as a thinking tool for intervention.

After excluding the participants who withdrew for illness or voluntary withdrawal, a total of 126 eligible participants were included in the analysis, distributed as follows: Control Group (N = 37), Experimental Group 1 (N = 41), and Experimental Group 2 (N = 48). The T-test revealed that the critical thinking skills of Experimental Group 1 were significantly higher than those of the Control Group ($p = 0.026$), but there was no significant difference between Experimental Group 2 and the Control Group ($p = 0.059$), nor between the two experimental groups ($p = 0.902$).

3.2. Experiment Design

A quasi-experiment was conducted using the GAI approach as the independent variable (one control group and two experimental groups). The dependent variable was the students' in-depth learning performance. The three classes all completed two periods of classes on information encoding, each lasting 40 min. As shown in Figure 1, students completed pre-tests including critical thinking disposition and skills before the classes. The first period, which was a lecture-style session, focused on the basic knowledge of information encoding, such as the definition, purpose, characteristics, and encoding rules. The second period involved the school uniform encoding scheme design task, in which the students completed the task collaboratively (4 per group) with communication with the teacher. The three classes adopted different teaching methods. After the experimental intervention, students completed an information encoding test indicating their in-depth learning performance.

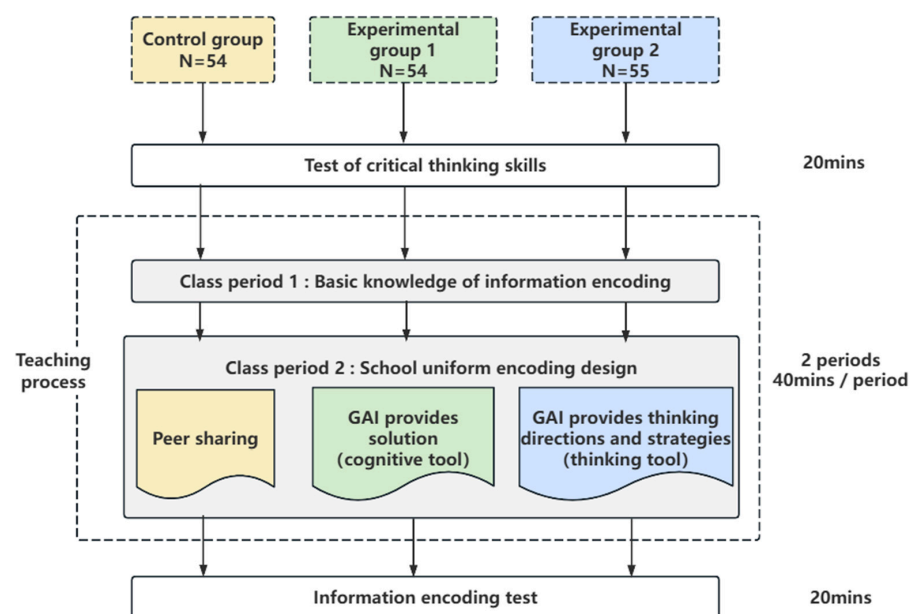


Figure 1. Experimental design.

To eliminate the effect of teacher teaching experience caused by the experimental order, the teacher gave the lessons in the order of Experimental Group 1, Experimental Group 2, and the Control Group.

3.3. Materials and Instruments

3.3.1. GAI-Generated Materials

Baidu's ERNIE Bot (<https://yiyan.baidu.com/>) was selected as the GAI tool for this study due to its free access, ease of use, and excellent user feedback in the Chinese context. The Control Group followed the conventional problem-based learning process in which they generated encoding schemes through peer sharing, and no GAI tool was

used. The two experimental groups used ERNIE Bot as a cognitive tool or a thinking tool to facilitate problem-based learning. Specifically, Experimental Group 1 used ERNIE Bot as a cognitive tool, where ERNIE Bot acted as an “assistant”, offering solutions for school uniform encoding for students to analyze, refine, and elaborate their schemes (Figure 2). Experimental Group 2 utilized GAI as a thinking tool, where ERNIE Bot acted as a “mentor”, demonstrating its thought process and strategies such as the “factor analysis” and the “goal analysis” for school uniform encoding to guide students’ thinking (Figure 3). Since sixth-grade students could not distinguish between cognitive tools and thinking tools and give appropriate prompt words to ERNIE Bot, in this study, the teacher designed prompt words for cognitive tools and thinking tools and asked ERNIE Bot to generate learning materials, which were printed and distributed to the corresponding experimental groups. In all three groups, the teacher selected and commented on typical student designs to help students optimize their designs.

3.3.2. In-Depth Learning Performance Measurement

To assess the students’ in-depth learning performance, an information encoding test was developed and used. The test comprises a retention section and a transfer section. The retention was measured by three single-choice questions, one multiple-choice question, and one yes/no question, covering the basic knowledge of encoding, such as its purpose, rules, and characteristics. The transfer was measured by an open-ended project design question requiring students’ knowledge application and transfer abilities across different task scenarios. Students must apply the encoding knowledge learned in class to solve the task scenario of “encoding electronic devices in the classroom”. Three experienced teachers reviewed the test to confirm its high relevance to the teaching content and suitability for sixth-grade students.

3.3.3. Prior Knowledge Test

Subject-specific knowledge tests or self-assessment questionnaires were typically used to measure prior knowledge. In this study, students’ prior knowledge was represented by the average of their scores in the mid-term exam of this semester and the final exam of last semester.

Please help me design a uniform coding scheme, providing specific coding rules and an example.

请你帮我设计一个校服编码方案，给出具体编码规则和示例

当设计一个校服编码方案时，可以考虑以下几个方面：

- 编码长度**：确定编码的长度，一般建议选择8-12位的编码长度，这样可以容纳足够的组合，同时也不会过长。
- 编码组成**：确定编码的组成方式，可以采用数字和字母的组合，或者只使用数字或字母。可以根据学校的特点和需求来选择合适的组成方式。
- 编码规则**：确定编码的规则，可以根据学生的年级、班级、性别等信息来进行编码。例如，可以将年级作为第一位编码，班级作为第二位编码，性别作为第三位编码，以此类推。
- 编码唯一性**：确保每个学生都有唯一的编码，避免重复。可以使用学号或者其他唯一标识符作为基础，再加上校服编码。
- 编码可读性**：尽量选择容易读取和记忆的编码方式，避免使用过于复杂或者难以理解的编码规则。
- 编码管理**：建立一个校服编码管理系统，用于记录和管理学生的校服编码信息。可以使用电子化的方式，方便查询和管理。
- 编码更新**：定期检查和更新校服编码，确保编码的准确性和有效性。

以下是一个具体的校服编码方案示例：

- 编码长度**：10位
- 编码组成**：数字和字母的组合
- 编码规则**：年级(2位)+班级(2位)+性别(1位)+顺序(5位)
- 示例编码**：20S001A
- 解释**：其中，“20”表示年级为20级，“S”表示班级为S班，“01”表示性别为男性，“A”表示顺序号为A开头。

请注意，以上只是一个示例方案，具体的校服编码方案需要根据学校的实际情况和需求进行调整和完善。

When designing a uniform coding scheme, the following aspects can be considered:

- Coding Length**: Determine the length of the code. It is generally recommended to choose a code length of 8-12 digits, which can accommodate enough combinations while not being too long.
- Coding Composition**: Determine the composition of the code. It can be a combination of numbers and letters, or just numbers or letters. The appropriate composition can be selected according to the characteristics and needs of the school.
- Coding Rules**: Establish the rules for coding, which can be based on students' grade, class, gender, and other information. For example, the grade can be the first digit of the code, the class the second, the gender the third, and so on.
- Coding Uniqueness**: Ensure that each student has a unique code to avoid repetition. Student ID numbers or other unique identifiers can be used as a basis, with the uniform code added.
- Coding Readability**: Choose a coding method that is easy to read and remember, avoiding overly complex or difficult-to-understand coding rules.
- Coding Management**: Establish a uniform coding management system to record and manage students' uniform coding information. An electronic approach can be used for convenient querying and management.
- Coding Update**: Regularly check and update the uniform codes to ensure their accuracy and validity.

Here is a specific example of a uniform coding scheme:

- Coding Length**: 10 digits
- Coding Composition**: A combination of numbers and letters
- Coding Rules**: Grade (2 digits) + Class (2 digits) + Gender (1 digit) + Serial Number (5 digits)
- Example Coding**: 20S001A
- Explanation**: Among them, "20" indicates the 20th grade, "S" indicates the S class, "01" indicates male gender, and "A" indicates the serial number starting with A.

Please note that the above is just an example scheme. The specific uniform coding scheme needs to be adjusted and improved according to the actual situation and needs of the school.

Figure 2. Examples of GAI as a cognitive tool.

If you were tasked with designing a uniform coding scheme, how would you approach the design process? Please present your specific thought process and strategic thinking.

如果你来设计一个校服编码方案，你会怎么思考来设计校服方案？请呈现你的思考过程和思维策略。

在设计校服编码方案时，可以运用因素分析和目标分析的思维策略来系统地解决问题。以下是如何使用这两个策略进行设计的详细说明：

因素分析：

- 收集信息**
 - 列出所有可能影响校服编码方案的因素，如学校文化、学生需求、成本预算、供应链能力等。
 - 进行市场调研，了解其他学校的编码实践和学生对校服的反馈。
- 评估影响因素**
 - 对每个因素进行重要性评估，确定哪些因素是关键性的，哪些是次要的。
 - 使用SWOT分析（优势、劣势、机会、威胁）来识别内部和外部环境的影响。
- 因素分类**
 - 将因素分为几大类，如“设计要求”、“功能性需求”、“用户体验”和“成本管理”。
 - 为每个分类的因素设定优先级。
- 制定策略**
 - 针对每组因素制定相应的策略，确保方案能够满足最关键的需求。
 - 考虑如何通过设计创新来平衡不同因素间的矛盾，例如成本与质量之间的平衡。

目标分析：

- 明确目标**
 - 确定校服编码方案的主要目标，如提高管理效率、减少库存误差、增强学生归属感。
 - 确保目标具体、可测量、可实现、相关性、时限性（SMART）。
- 细化目标**
 - 将主要目标分解为子目标，例如，提高管理效率可以分解为简化分发流程、加快库存查询速度等。
 - 确定每个子目标的优先级和实施难度。
- 目标策略匹配**
 - 将每个子目标与相应的策略相匹配，如使用条形码系统来加快库存查询速度。
 - 确保每个策略都与至少一个目标有明确的联系。
- 可行性分析**
 - 评估实现每个子目标的可行性，考虑资源、时间、技术等各方面的限制。
 - 根据分析结果调整目标或策略。
- 实施计划**
 - 为实现每个子目标制定详细的行动计划，包括时间表、责任分配、资源配置等。
 - 确保行动计划具有灵活性，能够适应可能出现的变化。
- 评估策略的有效性**
 - 正面影响：预期提升管理效率，减少人工操作失误。
 - 潜在挑战：技术支持和实施成本可能较高，但通过优化设计可降低支出。
 - 成本效益评估：确保方案的投入与学校实际需求相符，避免过度设计。

总结：
通过综合运用因素分析和目标分析的思维策略，可以设计出既符合学校实际需求，又具备高效性和可操作性的校服编码方案。

When designing a uniform coding scheme, one can utilize the thinking strategies of factor analysis and goal analysis to systematically solve problems. Below are detailed instructions on how to use these two strategies for design:

Factor Analysis:

- Information Collection**
 - List all factors that may affect the uniform coding scheme, such as school culture, student needs, cost budget, supply chain capabilities, etc.
 - Conduct market research to understand coding practices in other schools and student feedback on uniforms.
- Factor Evaluation**
 - Assess the importance of each factor to determine which are critical and which are secondary.
 - Use SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) to identify internal and external environmental impacts.
- Factor Classification**
 - Classify factors into several categories, such as "Design Requirements," "Functional Needs," "User Experience," and "Cost Management."
 - Set priorities for factors within each category.
- Strategy Formulation**
 - Develop strategies for each group of factors to ensure that the scheme meets the most critical needs.
 - Consider how to balance contradictions between different factors through design innovation, such as the balance between cost and quality.

Goal Analysis:

- Define Goals**
 - Establish the primary goals of the uniform coding scheme, such as improving management efficiency, reducing inventory errors, and enhancing student belongingness.
 - Ensure that goals are specific, measurable, achievable, relevant, and time-bound (SMART).
- Goal Refinement**
 - Decompose primary goals into sub-goals, for example, improving management efficiency can be broken down into simplifying distribution processes, accelerating inventory query speeds, etc.
 - Determine the priority and implementation difficulty of each sub-goal.
- Goal-Strategy Matching**
 - Match each sub-goal with the corresponding strategy, such as using a barcode system to speed up inventory queries.
 - Ensure that each strategy has a clear connection with at least one goal.
- Feasibility Analysis**
 - Assess the feasibility of achieving each sub-goal, considering resource, time, and technological constraints.
 - Adjust goals or strategies based on analysis results.
- Implementation Plan**
 - Develop detailed action plans for achieving each sub-goal, including timelines, responsibility assignments, resource allocations, etc.
 - Ensure that action plans are flexible and can adapt to potential changes.
- Strategy Effectiveness Evaluation**
 - Positive impacts: Expected improvements in management efficiency and reduction in human operation errors.
 - Potential challenges: Technical support and implementation costs may be high, but these can be reduced through optimized design.
 - Cost-benefit assessment: Ensure that the investment in the scheme aligns with the school's actual needs to avoid over-design.

Conclusion:
By comprehensively utilizing the thinking strategies of factor analysis and goal analysis, one can design a uniform coding scheme that not only meets the school's actual needs but also possesses high efficiency and operability.

Figure 3. Examples of GAI as a thinking tool.

3.3.4. Thinking Quality Test for Pupils (Short Version)

In this study, a short version of the Thinking Quality Test for Pupils (TQT4P-S) modified from The California Critical Thinking Skills Test (CCTST) was used to test participants' critical thinking skills. The TQT4P has a total of 25 questions, and a total of five sets of 5 questions each are used to test the five dimensions of critical thinking, which are reading comprehension, logical reasoning, fact judgment, hypothesis identification, and argument evaluation. Since reading comprehension is not a component of critical thinking, it was excluded from the test in this study, not due to internal consistency reasons, but because the original scale included it solely to study its impact on a paper-based critical thinking skills test, which proved to be unnecessary. At the same time, considering that the time for primary school students to complete the test should not be too long, we only selected three items for each of the remaining four dimensions. In this way, the short version of TQT4P we used had a total of 12 questions, with logical reasoning, fact judgment, hypothesis identification, and argument evaluation each being tested by 3 questions. Details on TQT4P-S and its reliability and validity are shown in Table S2 in the Supplementary Materials.

3.4. Data Analysis

SPSS 27.0 was utilized for descriptive statistics, one-way analysis of variance (ANOVA), and hierarchical multiple regression to identify the significant impact factors of in-depth learning performance and interactive effects among these factors. During hierarchical

multiple regression, variables were included in the analysis in order from stable to unstable. Specifically, critical thinking skills were first included, then prior knowledge, then use of GAI, and finally the interaction between critical thinking skills and use of GAI and the interaction between prior knowledge and use of GAI.

4. Results

4.1. Effect of Use of GAI on In-Depth Learning Performance

A one-way ANOVA was performed to examine the effect of GAI on students' in-depth learning performance (Table 3). No significant difference appeared among the three groups in the coding knowledge test ($F = 1.360$, $p = 0.260$). However, significant differences were observed in the transfer test ($F = 12.797$, $p = 0.001$). Both Experimental Group 1 (GAI as a cognitive tool) ($M = 4.329$) and Experimental Group 2 (GAI as a thinking tool) ($M = 3.510$) scored significantly higher than the Control Group ($M = 1.959$).

Table 3. Students in the experimental groups had similar performances in the coding knowledge test but significantly higher ones in the transfer test.

Variable	Group	N	M	SD	F	Sig.
Knowledge Retention	Control Group	37	21.380	3.419	1.360	0.260
	Experimental Group 1	41	21.340	3.490		
	Experimental Group 2	48	22.290	2.361		
Knowledge Transfer	Control Group	37	1.959	1.969	12.797	0.001
	Experimental Group 1	41	4.329	2.097		
	Experimental Group 2	48	3.510	2.177		

To further explore the differences in knowledge transfer performance, LSD post hoc tests were conducted. The results (see Table 4) showed that both Experimental Group 1 and Experimental Group 2 performed significantly higher than the Control Group, indicating that the introduction of GAI enhanced the students' in-depth learning performance. Yet, there was no notable distinction observed in the knowledge transfer performance between the two experimental groups.

Table 4. LSD post hoc test results: Experimental Group students outperformed Control Group in transfer test.

Variable	Group (I)	Group (J)	Mean Difference (I–J)	SD	Sig.	95% CI	
						Lower	Upper
Knowledge Transfer	Control Group	Experimental Group 1	−2.3698 **	0.4743	0	−3.309	−1.431
		Experimental Group 2	−1.551 *	0.4576	0.001	−2.457	−0.645
	Experimental Group 1	Control Group	2.3698 **	0.4743	0	1.431	3.309
		Experimental Group 2	0.8189	0.4449	0.068	−0.062	1.699
	Experimental Group 2	Control Group	1.551 *	0.4576	0.001	0.645	2.457
		Experimental Group 1	−0.8189	0.4449	0.068	−1.699	0.062

* $p < 0.05$, ** $p < 0.001$.

Considering there was no significant difference between the two experimental groups, the two experimental groups were merged into one in the following analysis and uniformly called the Experimental Group.

4.2. Factors Influencing In-Depth Learning Performance

In-depth learning performance was taken as the dependent variable, and critical thinking skills, prior knowledge, and the use of GAI were taken as the independent variables for hierarchical regression analysis (Table 5).

Table 5. Hierarchical regression analysis results of in-depth learning performance with critical thinking, prior knowledge, and GAI.

In-Depth Learning Performance	Model 1		Model 2		Model 3		Model 4		Model 5	
	β	t	β	t	β	t	β	t	β	t
Critical thinking skills	0.315 **	3.675	0.237 *	2.617	0.224 *	2.478	0.237 **	2.717	0.222 *	2.533
Prior knowledge			0.212 *	2.339	0.150	1.536	0.156	1.646	0.214 *	2.030
GAI					0.148	1.580	0.200 *	2.178	0.214 *	2.315
Critical thinking skills * GAI							0.260 **	3.142	0.230 **	2.661
Prior knowledge * GAI									0.123	1.237
R ²	0.099		0.138		0.155		0.219		0.229	
ΔR^2	0.092		0.123		0.134		0.193		0.197	
F	13.509 **		5.472 *		2.495		9.969 **		1.530	
VIF max	1.000		1.157		1.372		1.372		1.718	

* $p < 0.05$, ** $p < 0.001$.

When only critical thinking skills were included, according to Model 1, the standardized coefficient of critical thinking skills was 0.315 ($p < 0.001$), indicating a significant impact of critical thinking skills on in-depth learning performance. Model 1 was statistically significant ($F = 13.509$, $p < 0.01$) and explained 9.2% of the variance of students' in-depth learning performance.

After including prior knowledge as an independent variable, according to Model 2, the standardized coefficients of critical thinking skills and prior knowledge were 0.237 ($p < 0.05$) and 0.212 ($p < 0.05$), respectively, indicating significant impacts of critical thinking skills and prior knowledge on in-depth learning performance. Model 2 was significant ($F = 5.472$, $p < 0.05$) and explained 12.3% of the variance of students' in-depth learning performance.

However, according to Model 3, when GAI was introduced, only critical thinking skills had a positive and significant effect on in-depth learning performance ($\beta = 0.224$, $p < 0.05$). Meanwhile, Model 3 was not significant ($F = 2.495$, $p > 0.05$). The introduction of GAI has no significant positive effect on in-depth learning performance. However, with the introduction of various independent variables, the explanatory power of the model became stronger, and Model 3 explained 13.4% of the variance of students' in-depth learning performance.

According to the three models, when critical thinking skills, prior knowledge, and the use of GAI were introduced into the hierarchical multiple regression, ignoring their interactions, critical thinking skills were identified to have a positively significant effect on in-depth learning performance, while the use of GAI was not.

4.3. Interactive Effects Between Critical Thinking Skills, Prior Knowledge, and the Use of GAI on In-Depth Learning

The interactive effect of GAI and the other two variables (critical thinking skills and prior knowledge) was consequently analyzed by hierarchical regression. Model 4 added

the interaction item of critical thinking skills and the use of GAI based on Model 3, and Model 5 further added the interaction item of prior knowledge and the use of GAI. Model 4 was statistically significant ($F = 9.969, p < 0.01$) and explained 19.3% of the variance of students' in-depth learning performance. However, Model 5 was not significant ($F = 1.530, p > 0.05$). Therefore, we adopted Model 4 for further analysis in this study.

According to model 4, in addition to the contribution of critical thinking skills and the use of GAI (critical thinking skills: $\beta = 0.237, p < 0.01$; GAI: $\beta = 0.200, p < 0.05$), the interaction of critical thinking skills and GAI also had a significant positive impact on in-depth learning performance ($\beta = 0.260, p < 0.01$). However, the impact of prior knowledge disappeared ($\beta = 0.156, p > 0.05$), and the interaction between prior knowledge and the use of GAI had no significant effect either ($\beta = 0.260, p > 0.05$).

The interaction of GAI and critical thinking skills in predicting in-depth learning performance is presented in Figure 4. With the introduction of GAI, the in-depth learning performance of students with high critical thinking skills ($M + 1SD$) was significantly optimized (Table 6). When GAI is not introduced, critical thinking skills can also improve students' in-depth learning performance. However, when GAI is introduced, the role of critical thinking skills in enhancing students' in-depth learning performance is strengthened.

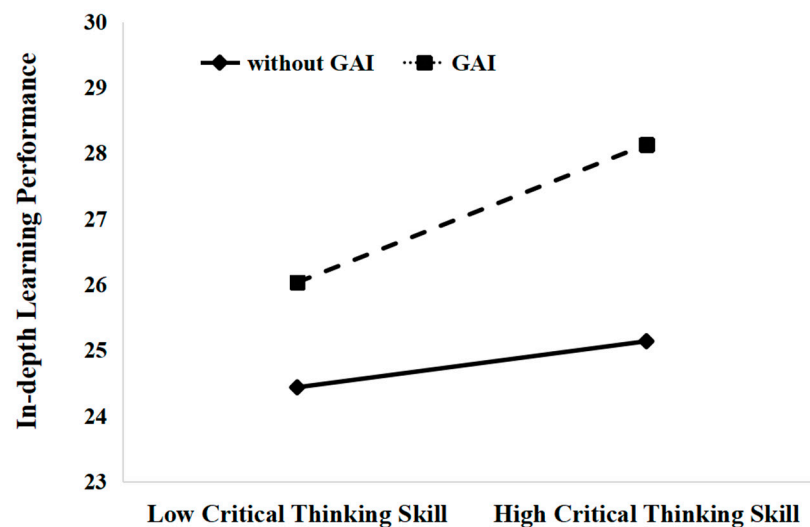


Figure 4. Interaction of GAI and critical thinking skills in predicting in-depth learning.

Table 6. Critical thinking skills total score in high and low groups.

Critical Thinking Skills	Effect	SE	t	p	LLCI	ULCI
−3.2496 (M − 1SD)	0.0271	0.9271	0.0292	0.9768	−1.8083	1.8624
0 (M)	2.2947	0.7468	3.0726	0.0026	0.8163	3.7732
3.2496 (M + 1SD)	4.5624	1.1379	4.0094	0.0001	2.3098	6.8151

5. Discussion

5.1. The Use of GAI Promotes In-Depth Learning

This study found that introducing GAI into the classroom could significantly improve students' in-depth learning performance. This can be explained by cognitive load theory. There are three types of cognitive load (Sweller, 2010): intrinsic cognitive load (ICL) imposed by the intrinsic nature of the material, external extraneous cognitive load (ECL) imposed by how the material is presented, and germane cognitive load (GCL) devoted to the processing, construction, and automation of schemas. These three kinds of cognitive

loads can be cumulative, but their total amount should not exceed the range of working memory when completing the same learning task (Sweller, 2010). Due to the limited cognitive resources of learners, the extraneous cognitive load (ECL) should be reduced as much as possible. At the same time, the GCL should be increased in the instructional design to promote the effective learning of learners. In this study, the introduction of GAI helped students reduce the ECL to some extent, freeing up more space for the germane cognitive load (GCL). In contrast, students in the Control Group merely discussed how to keep their uniforms from getting lost and make their labels unrelated to the class content, suffering from a heavier extraneous cognitive load (ECL).

5.2. GAI Amplified the Role of Critical Thinking Skills in In-Depth Learning

A finding of this study is that critical thinking skills and GAI jointly and interactively improved in-depth learning performance in a way that the use of GAI could amplify the in-depth learning effects for students with higher levels of critical thinking skills. This supports what was found in Yu and Wu's (2024) study, namely that ChatGPT should be applied as a tool for enhancing cognition. In this study, in-depth learning required the students to focus their cognition on whether the relevant factors of school uniform coding conform to the coding rules and characteristics, rather than using the GAI-provided answers directly. In this sense, using GAI-generated materials required a deep involvement of critical thinking.

Critical thinking skills are relatively complex higher-order cognitive skills; they cannot be applied mechanically or routinely but require judgment, analysis, and synthesis (Halpern, 1998). Some students were also concerned that solely relying on AI-generated content might lead to a loss of critical thinking skills and expected guidance on the critical utilization of GAI tools (Yuan et al., 2024). In this study, due to the unreliability of the accuracy of the information provided by GAI, judging the credibility and appropriateness of the GAI responses and deciding which to use and how to integrate into the problem-solving work put high requirements on students' critical thinking skills. Learners must consider various factors affecting credibility, including the authority, accuracy, and impartiality of the information source, in making judgments. Moreover, critical thinking skills are specific, actionable, and trainable abilities. They are unlikely to be developed solely due to individual growth but must be improved through experiencing a certain amount of learning or training (Kuhn, 1993). Moreover, according to the Monitor Hypothesis, learned grammar rules act as a monitor for checking and correct language output (Krashen, 1985). This study's finding that GAI enhanced in-depth learning when integrated with critical thinking and reduced reliance on prior knowledge could be related to this hypothesis. When students used GAI and engaged in critical thinking, it was akin to activating a "monitor" that helped them surpass their prior knowledge limitations. Instead of being constrained by their prior knowledge, they could utilize GAI to access new ideas, and through critical thinking, they processed and integrated this information to achieve in-depth learning. This process resembled the role of the monitor in Krashen's hypothesis, where the use of GAI and critical thinking as conscious learning could assist in enhancing learning outcomes by providing a more comprehensive and accurate understanding of what they learned. Therefore, students with higher-level critical thinking skills would have a more profound understanding of the materials provided by GAI and engage in a higher level of cognitive processing, thus amplifying their in-depth learning performance.

5.3. GAI Weakened the Role of Prior Knowledge in In-Depth Learning

Another interesting finding of this study is that the expected significant effect of prior knowledge on in-depth learning disappeared when the use of GAI was considered. In

traditional teaching and learning, according to Ausubel's theory of meaningful learning, prior knowledge plays a key role in the learning process, and the priority in instructional design is to figure out what students already know (Ausubel, 1963). However, this study found that in GAI-enhanced teaching, the difference in prior knowledge between students can be made up by learning materials generated by GAI, thus reducing the impact on subsequent learning. Combined with findings that the use of GAI amplified the role of critical thinking skills, this further revealed that learners' prior knowledge gap would be increasingly replaced by the gap in critical thinking.

5.4. Implication for Practice

This study can provide several implications for practice. First, GAI has excellent potential to facilitate in-depth learning, but whether it works depends on how it is used. Educators should facilitate students in using it as a cognitive or thinking tool. For example, students should be recommended to use GAI tools to generate learning materials or to elaborate their thoughts and to demonstrate thought processes and strategies for problem-solving. Second, teachers should focus on improving students' critical thinking skills when introducing GAI in learning, for they could amplify the effect of using GAI on in-depth learning performance. Specifically, teachers could guide and deepen students' thinking via encouraging them to analyze, evaluate, and refine the GAI responses, thus engaging them in critical thinking activities. Third, theoretically, the introduction of GAI shares low-order cognition and frees up more cognitive resources for higher-order cognitive processing, for which critical thinking skills are necessary. GAI could serve as a supplementary tool in school to provide required resources for students to make up their knowledge on learning or problem-solving. Teachers should also encourage students to use GAI tools to provide resources that are simpler to understand when they lack prior knowledge.

5.5. Limitations and Future Work

This study yields several limitations. Firstly, this study was conducted in October 2023, when GAI technology was still in its infancy. To prevent students from being exposed to inappropriately generated content, the study utilized materials that had been pre-generated by GAI. As a result, students were not able to interact with GAI during the learning process, which limited the thoroughness of the study's exploration of GAI's effects on in-depth learning and the interpretations we could make in this study. Secondly, the duration of the intervention was also relatively short, which might not fully exert the efficacy of the GAI designed as a thinking tool. That might produce no significant differences between using GAI as a cognitive tool and as a thinking tool. The same teacher taught the three groups in sequence, so it might also be challenging to eliminate the effect of teacher habituation. Thirdly, only the learning effectiveness of conceptual knowledge and transfer outcomes were measured, but they might not fully reflect all aspects of thinking development. Lastly, only 150 sixth-grade students from the same school were involved. The sample size might negatively impact the generalizability of this study because the various cognitive characteristics and needs of students of different ages were not considered.

Future research should further explore GAI's potential value and mechanism in thinking training by designing more comprehensive evaluation indicators and extending the intervention period. Future research can further delve into the cognitive characteristics and needs of students of different grades and design more targeted and adaptive ICT courses. Furthermore, future research should engage more participants from various backgrounds and test whether the same type of impact of ICT skills and prior knowledge would be found with other types of problems from other disciplines/subjects.

6. Conclusions

Through a quasi-experiment, this study explored the impact of using GAI, students' prior knowledge, and students' critical thinking on their in-depth learning performance in primary school Information and Communication Technology classes. The results suggested that students' critical thinking skills and the use of generative artificial intelligence significantly contributed to their in-depth learning while the impact of prior knowledge disappeared. Further analysis revealed significant interaction effects between the use of generative artificial intelligence and critical thinking skills, while no significant interaction was found between the use of generative artificial intelligence and students' prior knowledge. These findings revealed that GAI amplified in-depth learning performance for students with stronger critical thinking skills, while it possibly weakened the role of students' prior knowledge in in-depth learning. These findings help us to understand that the pivot for generative artificial intelligence to enhance learning is the critical thinking skills of learners, and the technology amplifies the role of critical thinking skills. Extending the current research that viewed critical thinking skills as the dependent variable that would be improved by using GAI and the ways to use it in an effective way (Habib et al., 2024; Korucu-Kiş, 2024; Liang & Wu, 2024), this study viewed it as a potential factor that might influence the effect of using GAI on improving in-depth learning performance. The findings of this study contribute to GAI research by extending its application to promote in-depth learning in the primary school context, which deepened the current research focused on improving teaching practice and some capabilities such as critical thinking skills (Korucu-Kiş, 2024; Liang & Wu, 2024). This study also contributes to the research in learning science by revealing the role of prior knowledge and critical thinking skills in using GAI to promote in-depth learning. This study highlighted how to design GAI as a cognitive or thinking tool to support in-depth learning, which might guide the orientation of applying GAI in classrooms.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/educsci15050554/s1>, Table S1: Detailed items of the Thinking Quality Test for Pupils (Short Version) (TQT4P-S); Table S2: Factor loadings, composite reliability, and internal consistency of the TQT4P-S.

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